

where σ is a uniform bound on the variance of the noise of gradient estimates, μ a lower bound on the objective function’s Hessian, $\mathbf{w}_0$ the initial point and $\mathbf{w}_*$ the optimum. For linear models, it is well-established that a larger LR accelerates the contraction of the bias term but has a detrimental impact on the variance term, ultimately leading the variance term to dominate. Coming back to the dynamics in Fig. 6, with a large ELR at the start, the convergence becomes primarily bottlenecked by the high variance term proportional to the learning rate, leading to higher loss values in the presence of WD. Conversely, towards the end of training, when ELR and the variance term are reduced, we see that WD catches up and performs better at the end, thanks to its relatively higher ELR *throughout the training* and thus better bias contraction. This perspective sheds light on the observation that EMA for LLMs is most advantageous when employed with large LRs (Sanyal et al., 2023) as we also illustrate in Fig. 23. As the variance dominates in this case, variance reduction of the averaging helps.

Experiments with bfloat16. Training in reduced precision is essential for speeding up training and reducing GPU memory requirements (Kalamkar et al., 2019). We further elaborate on the fact that WD is not fully equivalent to higher ELR and remains necessary for stable bfloat16 training. While Scao et al. (2022) observe that usage of float16 can cause loss divergences, bfloat16 is considered much more stable and is de-facto standard in LLM training. Although bfloat16 shares the same floating-point exponent size as float32 (thus, the *range* of possible values is the same), it offers lower precision, with only 7 bits for the fraction instead of 23. We observe that even more stable bfloat16 data type can still exhibit late-training spikes that irreparably harm model performance in *standard practical settings*, such as with a larger context length (e.g., 1024 instead of 256 as in the previous experiments). Therefore, we focus on this configuration for the experiments shown in Fig. 8. Runs with a moderate LR 0.001 (the default LR of Adam) without WD exhibit late-training divergence for *all* random seeds when using bfloat16. By comparison, training with float32 remains entirely stable. Importantly, we observe that the model *does not recover* after the loss spikes which contrasts with the loss spikes described in the Edge of Stability phenomenon (Cohen et al., 2021, 2022). We emphasize that all these runs use gradient clipping with the standard ℓ_2 -threshold. Finally, we observe that divergences can be prevented by reducing the LR, e.g., from 0.001 to 0.0006. However, this adjustment leads to slower training, as illustrated in Fig. 24 in the Appendix. Instead, the most effective approach is to use a higher LR of 0.001 *with WD*, which enables stable bfloat16 training and yields a better final training loss.

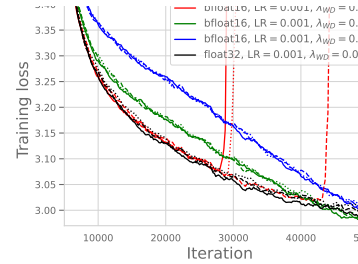


Figure 8: **GPT-2-124M on OpenWeb-Text with context length 1024.** Weight decay prevents divergence for LR 0.001 and enables stable bfloat16 training. The three random seeds are denoted with —, ---, ... lines.

4 Conclusions

In this paper, we demonstrate how weight decay, a single hyperparameter, can manifest three distinct effects across different training regimes: it offers regularization when combined with stochastic noise, improves optimization of the training loss, and guarantees stability in low-precision training environments. In the over-training regime, the scale of the noise $\sigma_{\eta,\lambda}$ is the fundamental quantity governing the implicit regularization strength of SGD. Weight decay combined with large LR enables the noisy dynamics to evolve by maintaining the scale at a constant level. Techniques such as EMA or fine-tuning work by reducing noise, thereby allowing for the effective exploitation of the accumulated hidden regularization. Coming to the under-training regime, AdamW (Loshchilov & Hutter, 2019) was introduced as a *regularization* method. Instead, we argue that it is effective as it modulates the ELR to attune the bias-variance tradeoff. In addition, it also improves the stability of training. In summary, weight decay is seldom valuable as an explicit regularizer; instead, its widespread adoption can be attributed to its ability to induce desirable changes in optimization dynamics. We also acknowledge limitations of our work: given our limited computational resources, we do not conduct truly large-scale experiments. Moreover, we do not prove new theoretical results. Instead, we strive to provide a clear experimental picture and formulate general explanations for the effectiveness of weight decay in different training regimes.

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